

Alexander Milner

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Degrees

PhD at Edinburgh University

School of Mathematics

September 2024 - July 2028

- Working with supervisor Dr Dimitra Kosta on a fully-funded studentship from the School of Mathematics.

MMath Master of Mathematics at Durham University

Department of Mathematical Sciences

September 2020 - July 2024

- Exam average of 92% with 92% in dissertation.
- **Prizes awarded:** Best exam results out of Durham Mathematics cohort in 2nd/3rd/4th years and best overall degree, winning the Margaret Millington award at graduation ceremony.

Publications

- [Distance Reduction in Bouquet Decompositions and Toric Ideals of Graphs](#) (pre-print) with Dimitra Kosta and Oliver Clarke, May 2026.
- [The Smallest Invariant Factor of Elliptic Curves, and Coincidences](#) (pre-print) with Jack Shotton, April 2026.
- [Computing all minimal Markov bases in Macaulay2](#) (pre-print) with Oliver Clarke, February 2025.

Talks

- Rigid Structures in Algebraic Combinatorics and Algebraic Statistics, May 2026.
- Hodge Club Seminar at Edinburgh University, May 2026.
- 7th Graduate Student Meeting in Applied Algebra and Combinatorics, April 2026.

Other research Experience

Research project funded by London Mathematical Society Undergraduate Research Bursary

- Under supervision of Dr Jack Shotton, studied the density of primes such that the $\mathbb{F}_p$ -rational group of points on an elliptic curve without complex multiplication has full j -torsion.

Masters Dissertation

- Under supervision of Dr Dan Evans, investigated applications of the Polynomial Method to Combinatorics.
- Extended results by Kezdy and Snevily and used the findings to prove a new result classifying when elements in a matrix can be rearranged to make said matrix invertible using commutative algebra.

Teaching Experience

University Tutorials

- Ran workshops for Algebra, Fundamentals of Pure Maths, Introduction to Number Theory and a masters level course in Python Programming at Edinburgh University.
- Taught weekly Calculus tutorials and marked first year assignments at Durham University.

Mathematics competitions

- Competed in 2023 Simon Marais Competition, coming 7th in the Pairs Division winning A\$1600.
- Competed in 2022 Simon Marais Competition, coming 9th in the Pairs Division winning A\$800.

Other skills and interests

- High level of proficiency with Python, Javascript, SageMath and Macaulay2.
- Keen conductor, violinist, pianist and organist, currently conducting Edinburgh University Sinfonia, having conducted the Durham University Symphony Orchestra from 2021-2024.